

Credit Risk Analytics

DataDNA Challenge | September 2025

Credit Portfolio Overview

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Default Risk Calculator

Portfolio Simulation



Loan Portfolio
312,4M



Borrowers
32,6K

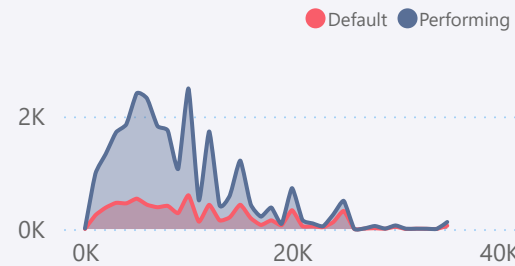


Default Rate
21,8%

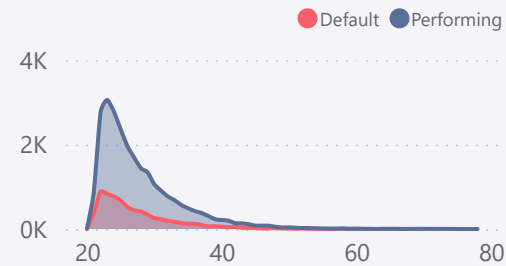


Portfolio at Risk
38,8M

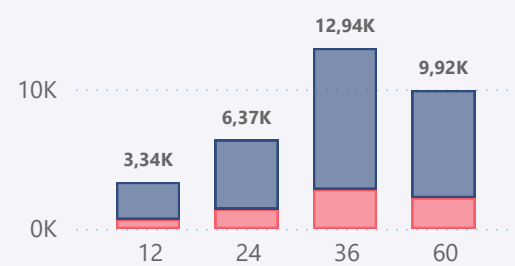
Loans by Amount



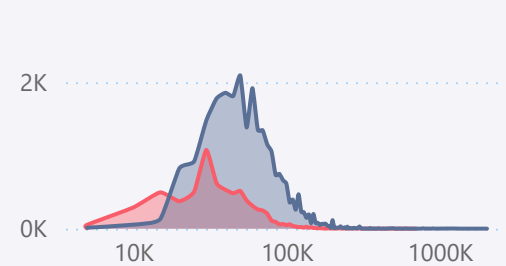
Borrowers Age



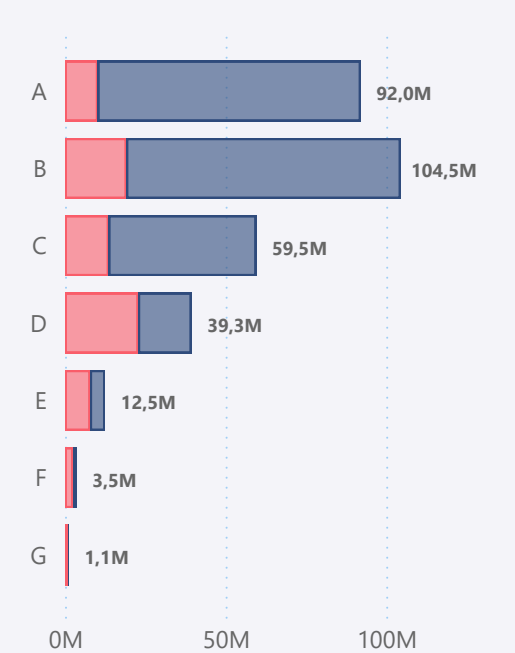
Loans by Term



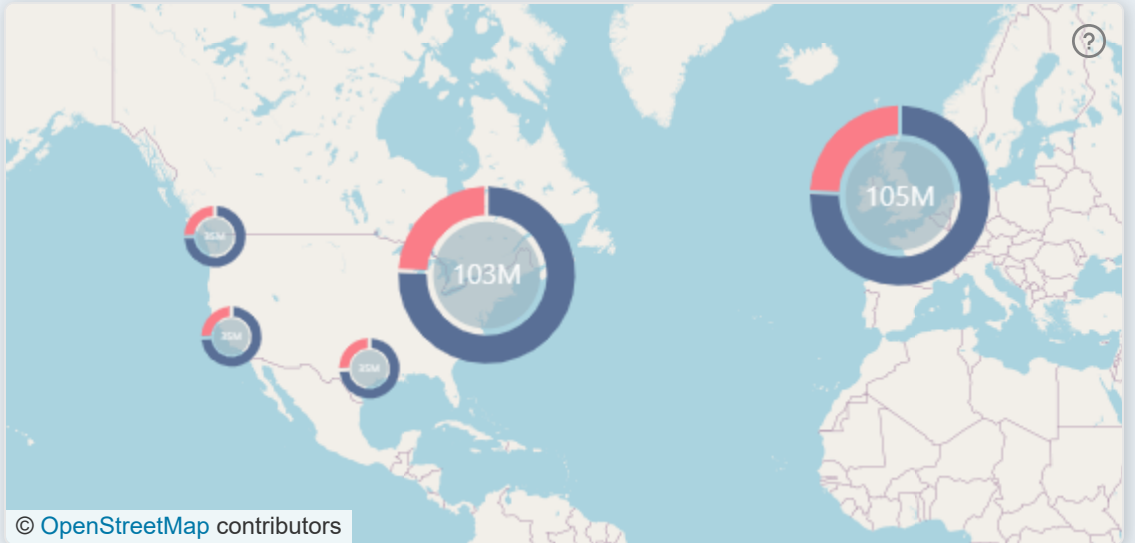
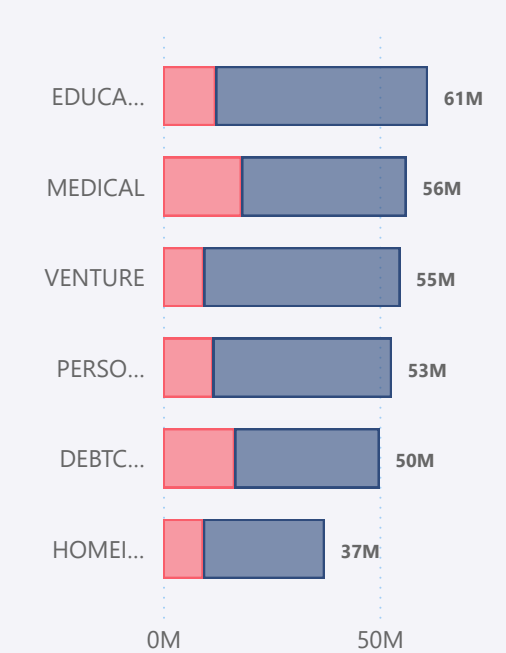
Borrowers Income



Loan Amount by Loan Grade



Loan Amount by Loan Intent

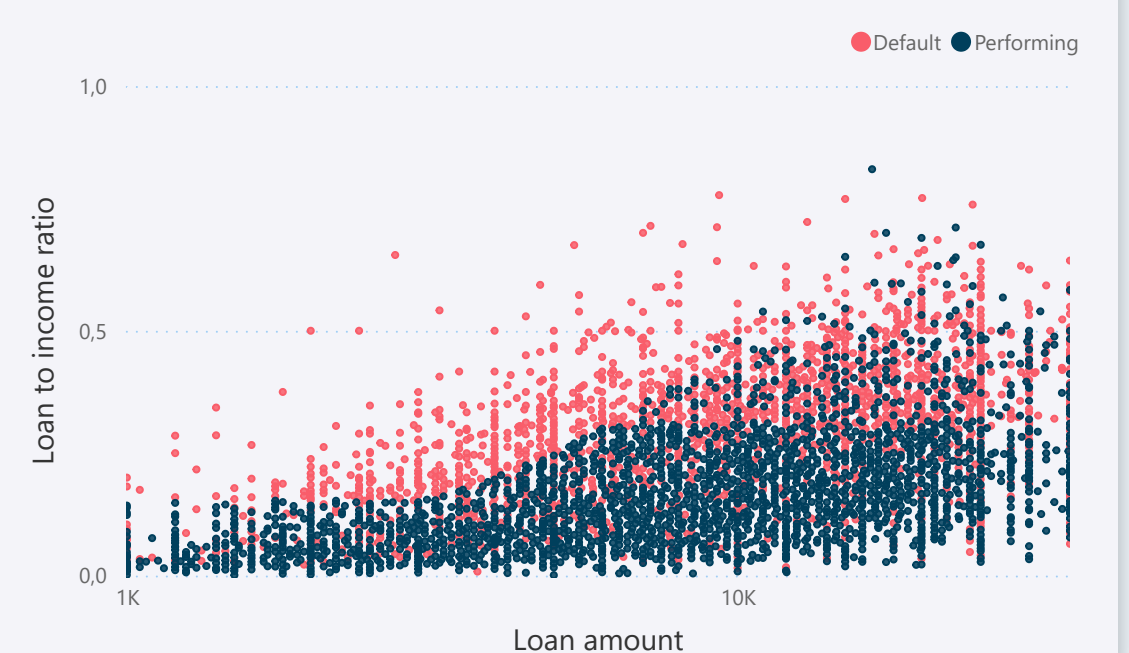


☐ Canada

☐ UK

☐ USA

Loan Status by LTI and Loan Amount



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Precision
0,8494



Recall
0,9572

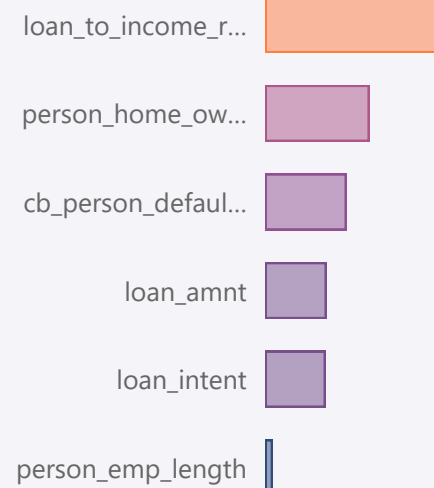


F1 score
0,9001



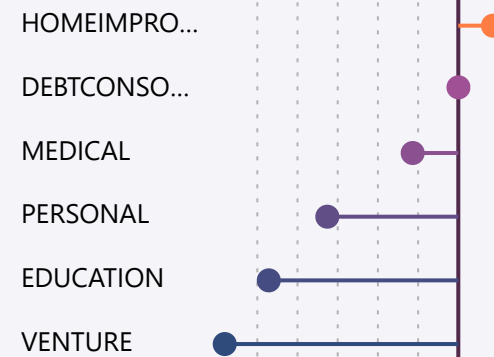
Balanced Accuracy
0,6798

Key factors

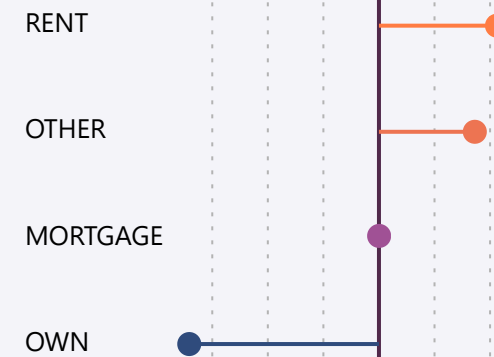


Marginal effects*

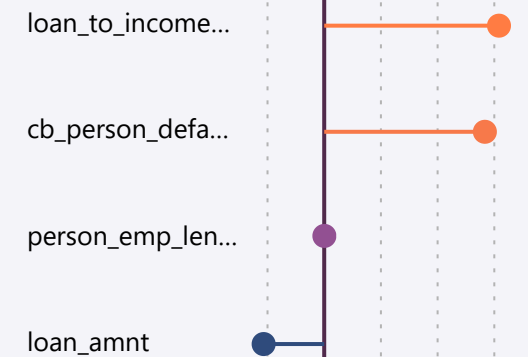
Loan intent



Home ownership



Other variables



* Standardized effects calculated using a Logistic Regression model

Default risk calculator*

Loan intent

- ☐ DEBTCONSOLIDATION
- ☐ EDUCATION
- ☐ HOMEIMPROVEMENT
- ☐ MEDICAL
- ☐ PERSONAL
- ☒ VENTURE

Home ownership

- ☐ MORTGAGE
- ☐ OTHER
- ☐ OWN
- ☒ RENT

Loan amount

9 500

Person income

67 000

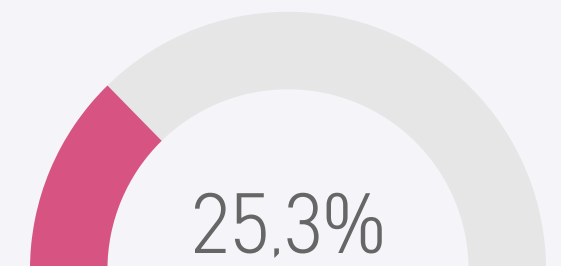
CB Default on File

- ☐ N
- ☒ Y

Employment length

5

Predicted PD



* Calculated using a Logistic Regression model

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Scenario Settings

Proportions

DEBTCONSOLIDATION

50

EDUCATION

50

HOMEIMPROVEMENT

50

MEDICAL

50

PERSONAL

50

VENTURE

50

Parameters

Borrowers Count

32 500

Bank Margin

25%

Recovery Rate

50%

Loan-to-income Cut Off

99%

Default-on-file Cut Off

0%

Rent and Other Cut Off

0%

Results Summary

Expected Portfolio Volume

311,5M

Expected Interest

35,2M

Expected Gross Margin

8,8M

Expected Default Rate

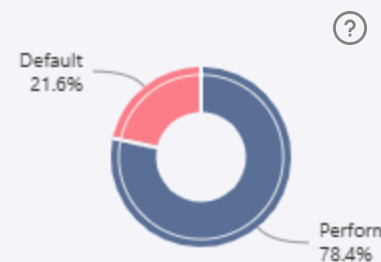
21,6%

Expected Loss

-18,5M

Expected Net Profit

-9,7M



Simulations are conducted on a synthetic dataset generated by sampling the original data with added jitter. The probability of default (PD) is predicted using an XGBoost model trained on the original dataset. The "Default-on-file" and "Rent and Other" cutoffs exclude a specified percentage of borrowers with the highest predicted PD.

Metric	Value
Balanced Accur...	0,8421
F1	0,9508
Precision	0,9205
Recall	0,9831
Sensitivity	0,9831
Specificity	0,7012

Loan Intent	Borrowers	Default Rate	Loan Portfolio	Interest	Gross Margin	Non-recov. Loss	Net Profit
DEBTCONSOLI...	5 199	28,6%	50,0M	5,6M	1,4M	-3,9M	-2,5M
EDUCATION	6 437	16,7%	60,6M	6,8M	1,7M	-2,9M	-1,2M
HOMEIMPROV...	3 596	26,3%	37,4M	4,3M	1,1M	-2,8M	-1,7M
MEDICAL	6 056	26,3%	56,5M	6,4M	1,6M	-4,0M	-2,4M
PERSONAL	5 507	19,8%	52,5M	5,9M	1,5M	-2,6M	-1,1M
VENTURE	5 705	14,7%	54,6M	6,1M	1,5M	-2,2M	-0,7M
Total	32 500	21,6%	311,5M	35,2M	8,8M	-18,5M	-9,7M

Calculated on a synthetic dataset with the original distributions